

CHAPTER

Unit-I



TECHNICAL COMMUNICATION

1.1 Communication

Meaning and Definition : The word **Communication** is of Latin origin, derived from the Latin word *communicare*, which means to share or exchange information, news and views etc. The Oxford Advanced Learner's Dictionary defines the word **communication** as the "activity or process of expressing ideas and feelings or of giving people information". In brief, communication can be called an act of transferring an idea or some news from one person to another.

While defining communication, Pauley and Riordan write, "*Communication is the act of transmitting an idea from one person to another. Communication always requires at least two people, the sender of the message and its receiver*". Here *sender* actually refers to the writer or speaker and *receiver* to reader or audience. According to Keith Davis, "*Communication is the transfer of information and understanding from one person to another person. It is a way of reaching others with facts, ideas, thoughts and values*". American Management Association defines communication as "*behaviour that results in an exchange of meaning*".

According to this definition **meaning** is central to communication. Thus, communication, in brief, is a process of sharing information.

1.2 Meaning and Definition of Technical Communication

The word 'technical', as defined by the Oxford Advanced Learner's Dictionary, refers to something "*connected with the practical use of machinery, methods etc. in science and industry*". Technical communication, therefore, means the transmission¹ of facts, figures, ideas and all sorts of scientific and technical information from one individual to another.

1. Passing something from one person to another.

The technical communication may be spoken as well as written. The technocrats² may conduct their professional jobs sometimes through oral communication which is performed through speech while sometimes written communication which depends on the written words becomes necessary. Hence, the technologist has to be proficient in both types of technical communication—oral and written.

All communication, general or technical, involves two persons in the process in which one is a speaker or a writer and the other is a listener or a reader. At the outset³, the speaker or a writer encodes a concept which he wants to communicate. At the second stage, that message or concept is decoded by the listener or the reader and converted into meaning. At this stage of decoding, the receiver analyses and interprets the meaning of the message so that any confusion or misunderstanding can be avoided. However, the technical communication can be internal or external.

1.3 Importance of Technical Communication

Technical communication gained momentum⁴ with the development of international trade and technology. In the seventeenth century English trade at international level progressed tremendously which resulted in increased international communication. The foundation of Royal Society in 1660 promoted the scientific researches and experimentations. The scientists need to prepare their reports and technically written articles. These articles were written in plain English and also in lucid and accurate style. The close of eighteenth century is marked by the outbreak of industrial revolution. The early nineteenth century witnessed an outgrowth of industries which also helped in bringing about a sort of revolution in the field of technical writing. Technical communication both oral as well as written came to the fore⁵ on account of the prevalent social conditions in the nineteenth century.

The present century is known for the growth of industries, science and technology. There is revolution in the field of communication technology and the scope of technical communication is very wide. Today we have innumerable national and international companies in our country. Consequently, technology has become the watch word of the present century. There is urgent need of fast means of communication. Today we have advanced means of communication such as computers, e-mail, internet, mobile phone etc. which help the technocrats in various fields. Technical communication—oral as well as written—has gained prominence in the wake of these modern means of communication. In the technological era of the new millennium⁶ also, it is important to get success in business and technology.

The present century witnesses a revolution in the field of the means of technical writing. The industrial revolution has effected significantly the means of communication technology. Earlier people used to document the proceedings of their research with the help of traditional means but computer, the innovation⁷ of modern times, has enabled technocrats now to prepare their technical reports, proposals, letters, memos, e-mail messages, professional summaries and so on more speedily and also more efficiently. These official documents help people to supply and receive information. Moreover, mobile phone and other means of information and communication technology which are the gifts of industrial revolution make professional interaction possible and also help people to reach to their professional and organizational goals.

2. Experts in Engineering; 3. Beginning; 4. Importance; 5. Take leading part; 6. A period of thousand years; 7. Discovery.

1.4 Meaning of General Communication

Communication is an integral part of every person's life. Effective communication, no doubt, is an important part of our professional life. Likewise communication is important for social relationships, international contacts when people verbally⁸ or nonverbally⁹, in speech or in writing share with other persons their views, interests, opinions and so on. This type of communication which is not specific is termed as general communication and as such it differs from technical communication. The table, mentioned below, highlights the features of technical and general communication.

General Communication	Technical Communication
General Content	Specific and technical content
Informal in style	Formal
Subjective	Objective
Not for specific audience	For specific audience
Oral as well as written	Mostly written as well as oral
No technical vocabulary	Technical vocabulary
Use of extravagant ¹⁰ words	Simple but effective

In most of the organizations technical communication takes place in written form. Some salient features of technical writing are mentioned below:

1.5 Features of Technical Writing

Written communications are a powerful medium in all walks of life. The scientists, engineers, researchers and doctors write reports and prescriptions which are different from general writing. In general writing, the writer often indulges in bombastic¹¹ style but in technical writing, he develops a clear and smooth style. Some important features of technical writing are mentioned below:

1. **Use of familiar and easy words** : In professional and technical writing, instead of the abstract words, familiar words should be used. Technical writing does not require verbose¹² style or circumlocution¹³. Hence, the writer should not use too many words where few would do since evasive¹⁴ talk will divert the attention of the writer as well as of the reader from the main point.
2. **Use of clear sentences** : The writer should place the main idea at the outset. He should follow normal word order – subject- verb-object. In this word order, the first sentence is the topic sentence in which the main idea is presented. Later, the writer develops the idea in clear sentences. His style is not superfluous or redundant¹⁵.

8. Relating to words; 9. Having use of no words; 10. Superfluous; 11. Pompous; 12. Using more words than are wanted; 13. Use of many words where few would do; 14. Seeking to evade; 15. Superfluous, unnecessary.

3. **Use of sentences of moderate length** : Experts are of the opinion that sentences should be of moderate length. Too much short or too much long sentences make the writings either too simple or too difficult. Generally, sentences of 12 to 25 words should be written.
4. **Use of active voice** : In the sentences written in active voice, the performer of the action is more emphasized than the receiver i.e., subject is more emphasized than the object. The active voice, therefore, helps the readers to comprehend ideas more easily. For example, while using active voice, we write – *I wrote a report* , and in passive voice the word order is as follows– *A report was written by me*. The first sentence where the subject acts, appeals to the reader more than the second sentence in which the subject is acted upon.
5. **Avoid redundant phrases** : Redundant phrases should be avoided. Pauley and Riordan give a list of redundant phrases and also concise word or phrase in place of them :

Table 1

Redundant Phrase	Concise Word or Phrase
due to the fact that	because
employed the use of	used
basic fundamentals	fundamentals
completely eliminate	eliminate
alternative choices	alternatives
actual experience	experience
connected together	connected
final result	result
prove conclusively	prove
in as few words as possible	concisely

6. **Avoid inaccurate expressions** : Precision and accuracy are two important features of a technical writing. Inaccurate expressions may lead to obscurity¹⁶ that will ultimately mar the purpose of a technical writing.
7. **Logical division of paragraphs** : In order to make a technical document clear and meaningful, thoughts should be properly arranged into different paragraphs. A paragraph is, generally, defined as a unit of thought hence a single idea must be conveyed in a paragraph. Just as the usage of the grammatical principles and the rules of punctuation are necessary for coherence¹⁷ in a professional document so is the arrangement of ideas in different paragraphs.

16. Making unclear; 17. Consistency.

1.6 Features of General Writing

Language is the most remarkable invention of man. It enables him to express his ideas and emotions emphatically. In the words of Balasubramaniam, "*Language has a very important social purpose, because it is mainly used for linguistic communication*". Man uses language for transmitting his ideas to his fellow beings in speech or in writing. He writes poems, essays, novels or other sorts of writings which are generally subjective type of writings. Below are listed some important features of general writings:

1. Its content may be general in nature whose basic purpose is to amuse the readers.
2. Literary/general writings are often cast in personal or subjective style.
3. The style of general writing is decorative and often includes bombastic words.
4. It is not always for a specific reader.

1.7 Difference between Technical Writing and General Writing

Since the dawn of civilization, man has been expressing his emotions and thoughts into words – oral and written. Letters, applications, essays, poems, novels etc. are some popular forms or genres of written expression. But the reports, letters, papers, articles, theses and some other writings are popular among the professionals such as doctors, engineers, scientists and the researchers. These writings are jointly known by a common name i.e. technical writings while literary and other types of writings are regarded as general writings. Technical writing and general writing are different from each other from various points of view. Some basic differences between technical writing and general writing are listed below :

1. Technical writing and general writing differ from each other on the basis of their purpose. General writings including poetry, novel, drama, short story and some other types of writings aim at amusing and inspiring the readers but the technical documents including report, resume, proposal and such other primarily aim at imparting information, some instructions and explanations.
2. The difference between technical writing and general writing is also with regard to style. General or literary writings are cast in personal or subjective style whereas technical or professional writings are supposed to be free from personal bias¹⁸. Objective or impersonal style is the hall mark of technical writing. General writing including fiction or other expressive writing may be highly imaginative, employing figurative word usage and symbolism or fictitious¹⁹ facts and characters. Technical document is always based on facts.
3. The style of technical writing is accurate and precise while that of general writing poetic, decorative and bombastic.
4. The language of technical writing should be simple but effective while a creative writer may indulge in extravagant and pompous choice of words.
5. Technical writing conducts business hence it is related to some specific profession while general writing is related to life in general or life as a whole.
6. A technical document differs from a general writing in terms of its vocabulary. Technical vocabulary is used in technical writing while in general writing familiar words or words of common use are generally included.

18. Prejudice; 19. Imaginary or false.

1.8 Language as a Tool of Communication

Language is the most remarkable tool that man has invented. It enables him to express his ideas and emotions emphatically. To quote T. Balasubramanian, "Language has a very important social purpose, because it is mainly used for linguistic communication". Man can communicate even without using a language. A little baby cries when he is hungry or he slaps his mother when he is angry. In both the above examples, communication takes place without the use of language. But man needs language since he has fine ideas to express. It is language which distinguishes man from the animal world. Thus, man has a genuine need of linguistic communication. It is an important aid for socialization for it fosters emity and brotherhood round the globe. People of different castes and creed, of different ethnic groups and also of different nationality can share their views with one another if they know one another's language.

A language can be used in two ways for the purposes of communication. It can be spoken or written i.e., we can communicate either using the spoken medium or the written medium. No doubt, speech medium is more popular medium of communication than the written medium. "In fact, it came centuries before writing in the history of any language community." Keeping in view both the mediums of communication, the Encyclopaedia Britannica Vol. 13 defines language as "a system of conventional, spoken or written symbols, by means of which human beings, as members of a social group and participants in its culture communicate".

1.9 Purpose/Functions of Communication

While defining communication Keith Davis compared it to a bridge of meaning which helps a person to cross a river of misunderstanding. In fact, this definition highlights the purpose of communication in one's personal and professional life. The following are the main objectives of communication:

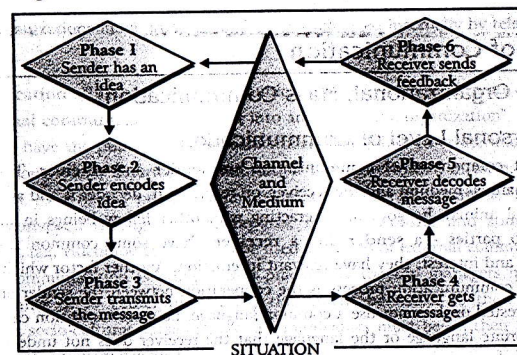
1. **Exchange of information** : The most important purpose of communication is to exchange information from one person to another. In business world, communication also occupies a prominent place since the employees receive different messages from the employer or the juniors from their seniors so that they can perform their job better.
2. **Persuasion** : One important purpose of communication is persuasion. Communication helps in persuading or convincing someone for a favourite response. Written as well as oral communication is used to influence others.
3. **Counselling** : Seniors counsel the subordinate staff from time to time whenever they face some difficulty at their work place.
4. **Co-ordination** : Communication is indeed an effective tool in co-ordinating the working of the employees of different departments or positions. Mutual understanding, joint consultation, seeking suggestions etc. are all possible through formal or informal communication. Besides, communication helps in releasing suffocating²⁰ emotions when one is able to know other's pent-up emotions. Finally, it leads to sharing ideas freely and in building up a tension free atmosphere.

20. Suppressed.

1.10 The Process of Communication

Communication doesn't occur haphazardly²¹ in organizations nor does it happen all at once. It is more than a single act. Communication is a dynamic, transactional (two-way process) that can be broken into six phases (See Figure mentioned below).

1. **The sender has an idea** : You conceive an idea and want to share it.
2. **The sender encodes the idea** : When you put your idea into a message that your receiver will understand, you are encoding it. You decide on the message's form (word, facial expression and gesture), length, organization, tone, and style-all of which depend on your idea, your audience, and your personal style or mood.
3. **The sender transmits²² the message** : To physically transmit your message to your receiver, you select a communication channel (verbal or non-verbal, spoken or written) and a medium (telephone, letter, memo, e-mail, report and face-to-face exchange). This choice depends on your message, your audience's location, your need for speed, and the formality required.
4. **The receiver gets the message** : For communication to occur, your receiver must first get the message. If you send a letter, your receiver has to read it before understanding it. If you're giving a speech, your listeners have to be able to hear you, and they have to be paying attention.
5. **The receiver decodes the message** : Your receiver must decode (absorb and understand) your message. The decoded message must then be stored in the receiver's mind. If all goes well, the receiver interprets your message correctly, assigning the same meaning to your words as you intended and responding in the desired way.
6. **The receiver sends feedback** : After decoding your message, the receiver responds in some way and signals that response to you. This feedback enables you to evaluate the effectiveness of your message : If your audience doesn't understand what you mean, you can tell by the response and refine your message.



21. Disorderly; 22. Passing something from one person to another.

As the above Figure illustrates, the communication process is repeated until both parties have finished expressing themselves. Moreover, effective business communicators don't cram²³ too much information into one message. Instead, they limit the content of a message to a specific subject and use this back-and-forth exchange to provide additional information of details in subsequent message.

Lesikar and Flatley in their book *Basic Business Communication*, to describe the communication process, use a situation involving two people Kelly and Justin. Communication takes place in the following steps :

- (i) **The Beginning : A Message sent :** The process begins when Kelly sends a message to Justin through a carefully selected medium or channel.
- (ii) **Entry in the sensory world :** The message enters Justin's sensory world. By *sensory world* Lesikar and Flatley mean all the noise that surrounds a person that the senses (sight, hearing, smell, taste, touch) can detect.
- (iii) **Detection by the senses :** From his sensory world Justin picks up stimuli (message) through his senses. Justin's senses detect message in accordance to the ability of his senses, his mental alertness, the distractions (noises) around him, his cultural background and above all his will. Message, thus detected by Justin's senses is sent to his brain.
- (iv) **The filtering process :** Kelly's message is filtered through Justin's unique mind and is given meaning. In other words, message is decoded by Justin according to his emotional make up, attitudes, beliefs, all cultural influences, his social groups and also his experiences. Thus, a message may not precisely give the same meaning what the receiver receives.
- (v) **Formation and sending of the Response :** Justin's mind reacts to the meaning, and he may respond. This response is unique (called feedback) since it is unique mind of Justin that forms it. This response may be through words, gestures, physical actions or some other means. Finally, Justin sends his message to Kelly. The same process can be repeated again and again when Kelly and Justin send messages to each other.

1.1 Levels of Communication

(Interpersonal, Organizational, Mass Communication)

1.1.1.1 Interpersonal Level of Communication

The most important components of communication are sender and receiver. The sender sends a message through a channel/ medium and the receiver, on receiving it, decodes it and arrives at a response. Since man is a social animal, he goes on interacting with other fellow beings in his day-to-day life. Whenever these two parties - a sender and a receiver have some common frame of reference, background, purpose and interest, they have relevant interaction. Another factor which can lead to some fruitful interpersonal communication process is the cooperation between the sender and the receiver. If one party is not interested or does not use a common language, no communication can take place. If the sender uses inappropriate language or the language that the receiver does not understand, his message fails to produce the desired response.

23. over-crowded.

The sender may transmit his message either through oral communication or written communication. In return he gets the response from the receiver. If the sender resorts²⁴ to oral communication either face-to-face or telephonic conversation, immediate response (feedback) is possible. In case of written communication, there is no immediate feedback. The most important thing for this interpersonal communication is clarity of purpose and skill in communicating the message.

1.1.1.2 Organizational Level of Communication

"Whether an organization is large, small, or virtual, sharing information among its parts and with the outside world is the glue that binds the organization together" (Bovee et al). In today's fast-paced e-commerce environment, organizations lay stress on team - work where the top officials and the low-level employees interact with each other. In such team based organization, communication becomes its life blood in which the information flows from down, up and across the organization.

In the wake of globalization and technological spurt²⁵, companies exchange messages with people outside the organization. Communication in an organization settings may be of two types :

- **Internal communication**
- **External communication**

Internal communication : Internal communication refers to exchange of ideas and information within an organization which may be formal or informal. Information may travel down, up, and across an organization's formal administrative hierarchy²⁶. Information, thus, is supposed to flow in :

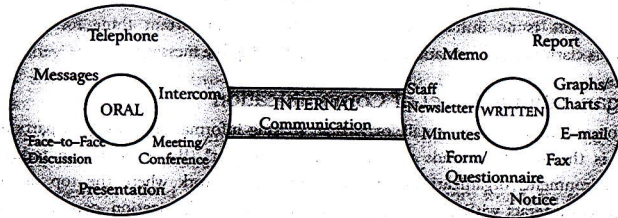
- **Downward flow**
- **Upward flow**
- **Horizontal flow**

The main motive of this internal communication is to help the employees to do their jobs. It helps to solve the problems or any complex situation and also develop a clear vision of the mission of the organization. In organizations much information can be exchanged internally by telephones, e-mail , fax, memo and so on.

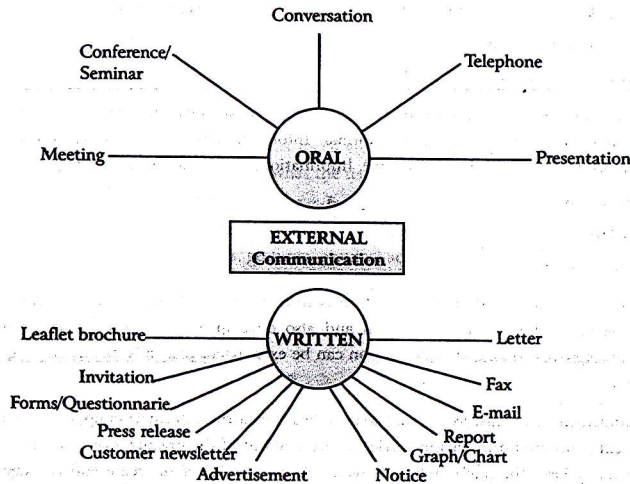
External communication : "Just as internal communication carries information up, down, and across the organization, external communication carries it into and out of the organization". Most of the business organizations today have the public relations team whose sole job is to create and manage the flow of formal messages to out-siders. External communication is usually formal in form. In order to create a favorable impression, good formal external communication becomes indispensable²⁷. Important messages to outsiders are conveyed through letters, reports, memos, website and sometime through oral presentation. Phone, fax, videotape etc. can also be used as means of external communication. Sometime, informal external communication is brought into practice by the top managers in order to know the needs of the customer or to gather information that might be useful to their organization. While interacting different socio-cultural outsiders, the communicators must be on their guard. They must be open-minded, tolerant and very courteous.

24. Sort again; 25. Sudden effort; 26. Gradation; 27. Essential.

Internal Communication



External Communication



[From Communication for Business by Shirley Taylor]

1.12 Mass Communication

Globalization and advances in technology have introduced people to the means of mass communication which can facilitate fast internal as well as external communication. Newspapers, magazines and periodicals are also used as means of written media. This type of communication is very effective. It is more persuasive and has a greater appeal. Technologies such as the internet, e-mail, voice-mail and faxes help people from all parts of the world to work together. These electronic forms of communication such

as audio-tape, teleconferencing and video-conferencing, closed circuit televisions increase options for communication. Below are mentioned some major electronic forms of mass communication.

Voice mail : Voice mail is a powerful tool of communication, used to replace short memos and phone calls.

Teleconferencing : It's an effective alternative for face-to-face meeting. It can be used for informational meetings.

Videotape : It is usually used for getting "motivational messages" out to a large number of people.

Faxing : It can be used to save time at the occasion when a hard copy of a written message is required.

E-mail : It offers a fast, low-cost means for communicating written brief message to a number of people at a time.

Internet messaging : Internet messaging is becoming a valuable business tool for receiving quicker response/feedback of your message, displayed on the sender's computer screen.

Web sites : Web sites offer "interactive through hyperlinks, allowing readers to absorb information consequentially".

1.13 The Flow of Communication

To succeed, an organization shares information with people both internally and externally. Each organization has a formal channel for the transmission of official messages. In today's fast paced e-commerce environment, organizations cannot limit themselves to the decision taken by the top officials, rather they depend on team work and collaborative²⁸ efforts in order to make the fast decisions in the global and competitive market place. To function in a team based organization information flows from upward, downward and horizontal in an organization's formal hierarchy. "The formal flow of information follows the official chain of command", comment Bovee et al. Whether the top managers or entry-level employees all need information to do their jobs. Managers and co-workers have to exchange the information and ideas to one another. In the network of formal organizational communication structure, there are three types of directional flow :

1. Downward flow
2. Upward flow
3. Horizontal flow or Lateral flow

1.13.1 Downward Flow

Main decisions in an organization are usually taken by the top officials and then they flow down to the subordinates or the co-workers who will carry out them. The information passes through written orders, rules, manuals and directives from the seniors to their juniors i.e., from higher level to the lower level in the organizational structure. For example, the President of an organization has Vice-President of Finance, Vice-President of Marketing, Vice-President of Research and Development, Vice-President of Production and so on. The orders flow from the President to these Vice-Presidents who further filter the orders to their respective departments. Such as Vice-President of Marketing communicates with Sales manager and Director of Advertising and Promotion, communication flows downward to special Project Supervisor and Advertising Production Chief.

● Purpose of Downward Communication

The purpose of the downward communication is to help the employees. In the words of Bovee et al, "Most of what filters downward is geared toward helping employees do their jobs. From top to bottom, each person must understand each message, apply it, and pass it along". According to Katz and Khan, the following are the five basic purposes :

- (i) To provide instructions how to do the job.
- (ii) To provide information which gives a rationale for the assigned job.
- (iii) To provide information about organizational policies and practices.
- (iv) To provide feedback on performance to the subordinates.
- (v) To provide information regarding the organization's mission or goals which the organization wants to achieve.

● Limitation of Downward Communication

When information flows from one level of an organization to the successive lower levels, the real spirit of message is lost. This process of filtration sometimes results in the loss of message.

Moreover, when the message is transmitted orally, the danger is that message may be distorted or miscommunicated. This is more when the message load is more than the handling in capacity of the lower level staff. Hence downward channels must be carefully used to avoid problems.

1.13.2 Upward Flow

To help the employees and make worthwhile decisions President/ Manager or a top official must be able to know what is going on in the organization. A Supervisor/Manager cannot be everywhere at once. An executive, therefore, has to depend on his subordinate staff that can furnish him information regarding different problems, grievances²⁹ and complaints of the lower level employees. Bovee, Thill and Schatzman cite an example to clarify this concept. They write, "As an employee, you are in a position to observe things that your supervisors and co-workers cannot see : a customer's first reaction to a product display, a supplier's brief hesitation before agreeing to a delivery date, an odd whirring noise in a piece of equipment, or a slow down in the flow of customers. Managers and co-workers need there little gems of information in order to do their jobs. If you don't pass that information along, nobody will—because nobody else knows. Needless to say that as the downward flow of communication is important so is the upward flow. Remember that upward flow is the opposite of downward flow.

Upward Communication includes the reports, proposals, suggestions, requests and some feed back. Upward flow of communication works when a Sales Manager submits his report to the Vice-President of Marketing, detailing the results of the survey conducted by the sales department.

Purposes of Upward Communication

The basic purposes of upward flow of communication are :

- (i) To develop faith/trust for the superiors in the hearts of the subordinates so that they can communicate freely.
- (ii) To provide a feedback.
- (iii) Sharing information of an organization helps to achieve organizational goals.

29. Real or fancied grounds of complaints.

Limitation of Upward Communication

The most important and apparent drawback of upward flow of communication system is that generally the subordinates highlight only the positive points of a survey i.e., they ignore the negative points which may lead to some problems in the long way. Moreover, the subordinates are reluctant to communicate upward, which turns into ineffectiveness of the upward flow of communication. Reason for this reluctance is the subordinate's fear that his report may displease his superior and he will miss his chance of promotion.

1.13.3 Horizontal/Lateral Flow³⁰

Communication in an organization also flows from one department to another either horizontally³⁰ or diagonally³¹. Since organizations use teams and collaborative work groups, horizontal communication becomes indispensable. When employees of the same work group, of the same level or of the same status share information with each other, in business world it is called as the horizontal or the lateral flow of communication. For example, the Special Project Supervisor and the Advertising Production Chief work in the same section—Advertising and Promotion Department. These two officials exchange their message and thus work smoothly. This horizontal flow of communication is termed as a 'bridge' or 'gangplank' by Henry Fayol.

Purposes of Horizontal Communication

The main purposes of horizontal communication system are :

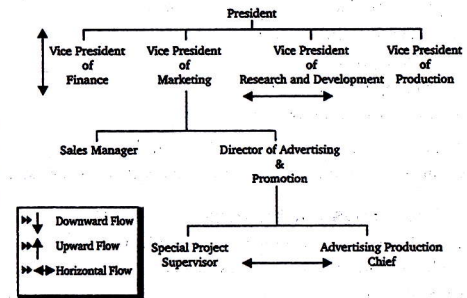
- (i) It helps employees to share information.
- (ii) It makes the co-ordination among peers³² possible.
- (iii) Whenever a department comes across with some problem, it may contact another department of the organization for its possible solution.
- (iv) This horizontal flow of communication is essential for effective functioning.

Limitation of Horizontal Communication:

Sometimes, this form of communication system creates some confusion. It can also lead to some conflict or clash among the peers. Misunderstanding and jealousy may be natural outcome of this channel of communication. This may also result in blocking continuous flow of information or in the congestion of modes of communication.

1.13.4 Spiral³³ or Diagonal³⁴ Communication

Another flow of communication is known as the spiral or diagonal flow in which people of different status and different departments share information in pursuit of achieving the organizational goals.



The Flow of Communication

30. Parallel; 31. Of a straight line; 32. Of the same status; 33. Circular; 34. Of a straight line.

1.14 Grapevine or Communication in Peer Groups

There is an informal communication network in every organization. This informal communication network, known as grapevine, carries information unofficially. It is an important source of information that supplements official channel. The employees at a work place often have casual conversations with their colleagues in an organization. Most of these conversations "deal with personal matters, about 80 percent of the information that travels along with grapevine pertains to business" (Bovee et al). This grapevine can be used to spread and receive informal messages.

1.15 Barriers to Communication

Communication does not occur haphazardly in organizations. It is a complex interactive process, a two way dynamic process which involves the sender and the receiver. Communication can be successful only when the receiver understands the message, sent by the sender. But if there are some errors and misunderstandings in communication, the message will be lost. There are some communication barriers which may block the communication process and thus prevent one from transmitting one's ideas meaningfully. No two individuals upon the earth are alike. We interpret the things according to our own perceptions³⁵, background and experience. When we communicate with a particular set of people, it becomes necessary to know about the background, education and capacity of the perception of the audience i.e., the receiver so that the message can be properly decoded. If a computer company representative gives the details of his company product to one who knows nothing about computer, in such a case, all his explanations and deliberations fail into insignificance since the receiver cannot perceive anything. Likewise if a doctor communicates the causes of cancer like fatal disease in strictly technical terms to a layman, the patient fails to understand. Doctor's wrong assumption about the patient's knowledge here creates a barrier to communications. There may be intrapersonal, interpersonal or organizational barriers which may hinder³⁶ the process of communication. While on the one hand, people's habits and attitudes block the communication process, the organizational barriers such as fear of superior, fear of getting transferred, distrust of communicator etc. hamper the communication. Bovee et al have listed some common barriers to communication in their book *Business Communication Today* which are listed below:

1. Perceptual and Language Differences
2. Restrictive Environments
3. Distractions
4. Deceptive Tactics
5. Communication Network

These negative forces affect the communication process. All these negative forces or barriers of communication may be categorized in another way as listed below:

1.15.1 Physical Barriers

The effectiveness of communication is affected by physical barriers like distance, noise or channel and the media used in the process. These physical barriers can be studied in some detail as follows:

(i) **Noise:** The success of communication depends on how the receiver receives the message, sent by the communicator. If any step in the communication process is blocked by some type of interference, the

35. Ideas; 36. Obstruct.

message is lost. Any interference in the communication process which distorts or obscures³⁷ the meaning of the message of the sender is generally called the 'noise'. Here the word 'noise' does not mean the mixture of loud unpleasant sounds but it refers to the barriers of communication which break the communication process.

Communication may be hindered due to the interference of someone or sometime due to the cracking voice on the telephone. In such situations it becomes difficult to convey message properly and also for the receiver to receive it properly. The message is, thus, partially conveyed or partially received, so will be the response. The barrier of noise may lead to misunderstanding and spoil business relations.

(ii) **Environment:** Physical barriers are often due to the nature of the environment. The natural barrier like the staff located in different buildings or on different sites, poor lighting or climate too cold or too hot affect the process of communication. The physical distractions like illegible³⁸ copy of the message, health problems may also hinder communication.

(iii) **Defects of Channel, Medium and Media (Communication Network):** *"Today's business people are plagued by message overload"* (Bovee et al). Modern technological developments have enabled the professionals to send and receive messages within no time. It encourages the employees to use as much as they can these fast means of communication such as e-mail, fax machines, voice mail, website, pagers and cell phones. Many a time, along with the crucial³⁹ and useful information, jokes and useless information are sent which ultimately result in overloading, causing congestion in network. Another reason which leads to overloading of information is the present hierarchical flow of communication which compels a lower-level employee to send his message through proper channel, i.e. there is no direct access from bottom to the top or from the top to bottom.

(iv) **Inadequate Facilities:** Lack of adequate facilities of communication such as phone, e-mail, fax etc. also works as barrier to effective communication.

1.15.2 Semantic Barriers

Semantics is the study of the meanings of words and phrases. The semantic barrier is related to the encoding and decoding of the message. Communication process involves two parties—the sender and the receiver whose mental images may not be identical. Here language plays a vital role. Since language is an arbitrary⁴⁰ code, the important thing is how two people can share the same meaning for a given word.

Bovee et al give an example to illustrate this. They choose a word *cookie*, for example. One may think of oatmeal, chocolate chip and sugar cookie but some other person may interpret the term cookie in its technological context—that is a computer file with information in it that is sent to the central server each time a particular person uses a *Network* or the internet. It concludes that two people may experience the same event but their perception may not be identical. Hence, for high degree of understanding, two individuals are to share perception and eventually share the meaning.

In a country like India where people have to communicate with culturally and linguistically diverse work groups, perceptual and language differences often hinder the effective communication. Moreover, as business today is crossing national boundaries to compete on a global scale, it has become necessary that

37. To make vague; 38. Difficult or impossible to read; 39. Important; 40. Not seeming to be based on some system or plan.

the employees must understand the language, the laws, customs and business practices of the people of other countries.

When some technical jargons are used, their misinterpretation also creates confusion and spoils all communication. For example, the word 'subject' in its ordinary sense means a 'course of study' while to a science student it stands for one on whom experiments are conducted.

1.15.3 Organizational Barriers

All internal factors which arise out of organization policies, culture, rules and regulations and obstruct the process of communication are called the **organizational barriers**. Some such factors are restrictive environment, deceptive tactics, lack of trust and closed mind set. Some of these factors are discussed below:

(i) **Restrictive Environments:** "The communication climate suffers when information is distorted, fragmented or blocked by an authoritarian style of management", comment Bovee et al. Sometime due to the authoritative style of management, the flow of information may be blocked either in upward, downward or horizontal direction. The people at the top of the organization prevent the lower-level employees from communicating their thoughts. The communication chain thus moves down but not up. Sometime a lack of effective means for sharing the idea may also block the communication process.

(ii) **Deceptive Tactics:** Another communication barrier which may distort the message is the use to the deceptive tactics by some manipulative communicators. Such communicators may hide the facts. Instead, they, for their personal gains, may exaggerated benefits and quote inaccurate data. Such deceptive communication tactics are known as communication barriers.

(iii) **Sense of Insecurity:** Lack of co-operation among the staff hinders the communication process. It becomes serious when the fear of transfer or demotion prevents the subordinate staff to give suggestions or raise fingers to some malpractices in an organization.

(iv) **Rigid⁴¹ Rules and Regulations:** No doubt rules and regulations help in the smooth working of an organization. But sometime too rigid rules work as barriers to communication. For example, the complex hierarchical structure obstructs the free flow of communication because in such a system a junior staff member has no direct access to the top officials. Consequently when the message reaches to the top official after passing through different other officials it either loses its basic form or is diluted.

(v) **Closed Communication Climate:** The status and power often hinders the effectiveness of communication. In an organization where the superiors are of closed mind set, do not have faith in their junior staff and always maintain difference in status, there can never be congenial⁴² atmosphere for healthy and free communication.

1.15.4 Socio-Psychological Barriers

Actually effective communication takes place when sender and receiver of the message have a favourable mind set towards each other. People's attitudes, their state of mind, their emotions and feelings may function as detractors⁴³. These interpersonal and intrapersonal barriers block the process of communication. Some socio-psychological barriers are listed below:

(i) **Attitudes:** The personal attitudes of individual employees which may be due to lack of motivation or interest at a particular work may obstruct effective communication.

41. Strict; 42. Suitable for work; 43. Diversions.

(ii) **Emotions/Moods:** If the sender is in bad mood or emotionally disturbed, he/she is not able to send the message properly nor he/she will receive other's message positively.

(iii) **Resistance to Change:** Sometimes the receiver makes a world of his/her own. He/she does not pay proper attention to the sender's message. Such a fellow is never open to accept other's logics or arguments. He/she either does not respond properly to other's communication or misinterprets the sender's message.

Besides these, other social barriers like closed mind set, difference in status, restrictive environment have already been discussed.

6 Gateways to Communication

There are a few guidelines for overcoming communication barriers. These guidelines are listed by Bovee et al in their book for making communication effective:

- 1. Adopt an Audience-Centered Approach :** Adopting an audience-centered approach means focusing on audience/receiver in order to make a meaningful communication. The communicator should try to know about the education, age, status and other relevant particulars of the receiver's personality. A message, written while keeping in view the receiver, may be understood and accepted easily.
- 2. Foster an Open Communication Climate :** As discussed earlier, restricted environment in an organization leads to misunderstanding and sometime distorts the meaning of the message. Hence communication should flow freely in all directions upward, downward and lateral. In order to foster open communication climate the number of the organizational levels should be modified. There should be fewer links in the communication chain. If the low-level employees have a direct access to the Manager, it will be a blessing in two ways-first it will help the employees to communicate their message directly without any distortion⁴⁴, and secondly it will enable the managers to share information with other employees and also seek their help in taking some decisions or solving some problems. It will also facilitate feedback by asking questions and encouraging the audience to express his reactions.
- 3. Commit to Ethical Communication :** Ethical communication means to observe transparency⁴⁵ in working. No deceptive tactics are to be used for profit. Even negative information is to be made public and graphic data presented fairly.
- 4. Create Lean, Efficient Messages :** If the useful and useless information are put together, they result in the audience's loss of interest and he fails in concentrating upon the useful message. In order to avoid overloading of information on communication network, some suggestions mentioned below, should be taken into consideration:

- The number of the messages should be reduced. It is better if fewer messages are sent.
- To make efficient messages, avoid jokes and chitchats.
- Distractions - physical as well as emotional - can be avoided
- People with good communication skills also create efficient messages. Hence master communication skills.

44. Manipulation; 45. Openness.

Questions and Answers

Q. 1. What is communication ?

Ans : Before we begin to practise technical writing, we must understand certain basic concepts such as the terms *communication*, *medium*, *document* and *generate*.

The word communication means share or exchange information, news etc. While defining communication, Pauley and Riordan write, "Communication is the act of transmitting an idea from one person to another. Communication always requires at least two people, the sender of the message and its receiver". Sender actually refers to the writer and receiver to the reader or audience.

The writer communicates the information in words to the reader or audience who gleans⁴⁶ the message by decoding the words. In the words of Green, "The document itself becomes in effect a blueprint from which the reader recreates the message".

Q. 2. What is the need of communication ? or What is a medium ?

Ans : There is no device through which the writer's mind can be read. Hence, the writer or sender sends a message through a medium. The writer records or 'encodes' his message in a common medium and then transmits it to the receiver. The receiver 'decodes' or tries to understand the message. Thus, a medium is defined as "a thing in the middle" by which the sender conveys ideas to the receiver.

Q. 3. Explain in brief the *process of communication*.

Ans : Communication is a two way process which can be broken into six phases:

1. The sender has an idea.
2. The sender encodes the idea.
3. The sender transmits the message.
4. The receiver gets the message.
5. The receiver decodes the message.
6. The receiver sends feedback.

Phase 1	Phase 2	Phase 3	Phase 4	Phase 5	Phase 6
Sender has an idea.	Sender encodes the idea.	Sender transmits the message.	Receiver gets message.	Receiver decodes message.	Receiver sends feedback.

Process of Communication

Q.4. What does the technical communication mean ?

Ans. The word technical as defined by the Oxford Advanced Learner's Dictionary, refers to something "connected with the practical use of machinery, methods etc. in science and industry." Technical communication, therefore, means the transmission of facts, figures, ideas and all sorts of scientific and technical information from one individual to another.

⁴⁶. To obtain information.

Q.5. What is the need of Technical Communication ? Mention main features of Technical Communication. (M.C.A., 2008-2009; B.Tech., SEM. I, 2009-10 & 2012-13 G.B.T.U., Lucknow)

Ans. Technical communication gained momentum with the development of international trade and technology. Today there is immense growth of industries, science and technology.

Technical communication has become the essence of the organizational life. There are innumerable national and international companies in our country. Consequently technology has become the watchword of the present century. In an organization, technocrats prepare reports, proposals, letters, memos and professional summaries. All this official documentation helps people to supply and also receive information. Thus, technical communication both oral as well as written has become the need of the hour of the modern industrial world as it helps people to reach⁴⁷ to their professional and organizational goals.

Technical communication differs from general communication. Communication which is not specific is termed as general communication. The table, mentioned below, highlights the special features of technical communication and also its differences from general communication :

Technical Communication	General Communication
Specific and Technical content	General Content
Formal in style	Informal
Objective	Subjective
For specific audience	Not for specific audience
Mostly written as well as oral	Oral as well as written
Technical vocabulary	No Technical vocabulary

Q.6: What do we mean by 'Technical Written Communication' ?

Ans : "Technical Written Communication is the practical writing that people do as part of their jobs". (Pauley and Riordan) Today all technocrats, whatever their position – from executive to manager – issue documents to their superiors and also to their juniors. It is a part of their responsibility. These documents enable them "to fulfill their roles in organization". In governmental units, schools, and hospitals, people need technical writing in order to maintain their operations. Hence, a technologist must achieve the communicative skill from the very beginning. Good technical writing is not a God gift. The meticulous⁴⁷ efforts of the learner may result in picking up good technical writing style.

Q.7. Write in brief the purposes of technical written communication ?

Ans : Killingsworth describes the three basic purposes of technical written communication : to inform, to instruct and to persuade. The first and foremost purpose of all technical writing is to inform. In government or semi-government offices, schools and hospitals, people have to supply and also receive information. A researcher should know how to prepare the documents, giving the information about the result of the experiment. Similarly a mechanical engineer has to furnish the information about the operation of a new machine.

⁴⁷. Paying careful attention to every detail.

Second purpose of technical written communication is to instruct. The technical writers give instructions for using equipment and also for performing duties. Such as the consumers are instructed how to use a new product, the medical practitioners are directed how to use a medicine.

Lastly, technical written communication aims at persuading the readers. A technical writer, for instance, may prepare a document to persuade readers to do certain things.

Q.8. Point out the basic difference between Technical Writing and General Writing.

(B.Tech., 2004-2005; B.Tech. SEM. I, 2011-12, M.T.U., Noida;
B.Tech., SEM. I, 2009-10, G.B.T.U. Lucknow)

Ans : Technical Writing differs much from general writing. The scientists, engineers, researchers and doctors make use of technical writing which needs precision and accuracy of expression. A scientist like a poet or a novelist cannot indulge in exaggeration and use of superfluous⁴⁸ words. His style cannot be bombastic. His style must be based on facts and he should use technical or special words, instead of general or common words. Like a general writer, a technical writer does not indulge in flights of fantasy⁴⁹ and poetic expression nor a technical writer can ever use figurative language. In a technical writing objective language is always used which brings certainty, avoiding all ambiguities. A business letter, a scientist's report and a document written by an engineer belong to the category of scientific and technical writing whereas a personal letter, a fictional tale, a speech on some general or common topic and also friendly conversations are known as the general writings.

Q.9. Write 150 words on the style of Technical Written Communication.

Ans : A clear and smooth style should be developed in order to communicate well through a technical writing. This clear and smooth style can be achieved after much practice. Since 'Practice makes a man perfect', the writer prepares the first draft and then revises and re-revises it.

Certain tips to achieve a good style are given below :

1. First of all, the writer must write clear sentences for his readers. For it, he should place the main idea first. He should use normal word order – subject- verb- object. Under this normal word order, the topic is revealed first and then the idea is developed. Such an order usually produces most concise sentences. Besides, the active voice should be used because it emphasizes the performer of the action rather than the receiver. In some special cases, the sentences in passive voice can also be used.
2. Secondly, sentences of 12 to 25 words should be used because too much short or too much long sentences make the writings either too simple or too complicated respectively⁵⁰.
3. Redundant phrases should be avoided, such as use of 'due to the fact that' for 'because'.
4. Noun clusters should be avoided. When three or more nouns are joined in a phrase, they make reading difficult.
5. Inaccurate expressions should also be avoided.
6. Instead of the abstract words, familiar words should be used.

Thus, a good writer should keep a restraint⁵¹ in style so that his language will not be ambiguous⁵² and it will be easily understood by every reader.

48. Unnecessary; 49. Imagination; 50. In a sequence; 51. Check; 52. Not clear.

Q.10. How does the purpose of technical writing differ from the purpose of general writing ? Illustrate with an example.

Ans : There are many types of general writing including fiction, expressive writing, expository writing and persuasive writing. Fiction that includes poetry, novel, drama and short story is highly imaginative that employs figurative word usage, symbolism and fictitious facts and characters to present a message. Expressive writing expresses the writer's personal feelings about a personal experience such as a person, a place, an event etc. Expository writing reveals the writer's knowledge about a subject. Persuasive writing includes the editorials in which the editor aims at controlling the emotional attitude of audience toward a topic.

Technical writing is different from general writing. "Technical writing requires give-and-take, dialogue, a follow-up and/or input and action." (Sharon J. Gerson/Steven M. Gerson). In general writing the writer does not expect any response from the reader. He does not receive any feed-back on his writing from his readers. But in technical writing, the detachment⁵³ between reader and writer does not exist. The purpose of technical correspondence is to link the writer and his superior – his supervisor. It also establishes a clear and direct link between the writer and his client or the writer and his juniors or his co-workers. Technical writing, in fact, creates action. It makes the reader to respond. For example, if a person writes a memo, he expects a reaction. This reaction may be in the form of some suggestion or job completed or problem solved. Likewise, if the writer writes a business letter, he expects to make a sale or to receive an answer to an inquiry. If a technical writer writes instructions, he knows that someone will follow the steps he has outlined. But a general writer writes in vacuum. He does not receive any immediate feedback.

Q.11. Write in brief about the importance or the relevance of Technical Written Communication.

or

What do you mean by purpose of the technical writing?

[B.Tech. SEM. II, 2011-12, G.B.T.U. Lucknow]

Ans : Technical writing is a unique form of writing which is not taken just for pleasure or entertainment. It is intended to communicate to a specific audience for a specific purpose. Technical writing is important to succeed in business and technology. It helps in conducting business. Below are given certain reasons which make technical writing a significant factor of a writer's work experience :

1. **Technical writing conducts business** : It is evident that technical writing is a major component of the work environment. Through technical correspondence instructions are maintained, technical description and specifications are prepared, summaries and resumes are prepared and sites on the World Wide Web are prepared which inform the world about one's company's product.
2. **Technical writing takes time** : Technical writing is time-consuming. The supervisors not only spend their time in writing but they also spend additional time in reviewing and revising the writing composed by their subordinates.
3. **Technical writing costs money** : Company pays its employees for their writing skill. Suppose an employee spends one hour a day in writing, it means company pays the employee for seven hours a week, time spent by him in technical writing. Good writing is supposed to have additional monetary values other than writer's salary or the cost of correspondence. Besides, good proposals generate income.
4. **Technical writing as an extension of one's interpersonal communication skills** : Technical correspondence is not merely writing a status report, a memo, a business letter or a resume but it reveals the writer to his readers. Your writing conveys to the readers your power to think, or think logically.

53. Not being involved.

Thus, good technical writing can accomplish more than just getting the job done. It tells the readers of the writer's competence and also helps the writer to sell well his communication skill.

Q.12. Give the characteristics of technical writing.

(B.Tech., 2004-2005)

Ans : Cunningham and Hartley have given four characteristics of technical writing which are as follow :

- It requires specific audience,
- It uses plain and objective language,
- It stresses presentational organization and
- It employs visual aids.

Specific Audience : Technical writing engages a specific audience. In any writing situation, audience has a specific role. A good writer prepares his documents keeping in view different audience roles. For example, a manual for operating a computer is prepared for the operator where the entire process is described.

Objective Language : Plain and objective language is a pre-requisite of a technical writing. The writer uses such language which the reader easily follows. No doubt, the writer uses the technical vocabulary but he presents the subject in plain and objective language. The purpose of a technical writer is to focus the readers' attention, to furnish information, to instruct and to persuade the readers.

Presentational Organization : Good technical writers employ such devices which enable them to make the format of document easy to scan. They often use repetition and white spaces between lines to emphasize important points. Such devices are used by the writers to make their presentational writing easy to glance at quickly.

Visual Aids : Visual aids are an essential part of technical writing. The writers use bar graphs, line graphs, pie charts etc., in order to communicate effectively. Technical writers use visual aids for four purposes : (a) to summarize data (b) to give readers an opportunity to explore data (c) to provide a different entry point into the discussion and (d) to engage reader's expectations.

Q.13. Why clarity and simplicity are the governing features of Technical Written Communication ?

Or

With a specific example, illustrate how clarity can be introduced in technical writing ?

(B.Tech., SEM. II, 2012-13, G.B.T.U. Lucknow)

Ans : Sharon J. Gerson and Steven M. Gerson write in detail about the importance of clarity and simplicity in Technical Writing. Technical Writing, they think, must be clearly worded and developed to avoid confusing its audience. They write –

"The ultimate goal of good technical writing is clarity. If you write a memo, letter, or report that is unclear to your readers, which your readers can't understand, then what have you accomplished? You have wasted time."

An unclear report fails to communicate the message to the reader. He does not understand the writer's train of thought. He then is forced to write an inquiry to the writer. The writer rewrites, clarifying his intentions. He has to write twice to accomplish the same goal. This is nothing but a sheer⁵⁴ wastage of time. Hence, in order to avoid these time-consuming endeavours, the writer should write clearly.

In order to achieve clarity, the writer must confine himself to supplying specific information. Vague and indefinite expression or use of unclear adjectives obscure the meaning of the writing. Another key to

clarity is using words which the readers can understand easily. Acronyms⁵⁵, abbreviations and jargon should not be used. Another important point to clarity is using verbs in active voice.

Q.14. Write a short note on the language of Technical Written Communication.

Ans : The ultimate goal of good technical writing is clarity and this can be obtained by using words which the readers can understand easily. Obscure words should be avoided. Old fashioned words or out-of-date terms which are difficult to understand should be avoided. "A modern thrust in technical writing is to write the way you speak. unless you speak poorly".

Following are certain tips for a writer of technical correspondence:

- A redundancy (that is a repetition) should be avoided.
- Pompous words or phrases should not be used as they may detract from the simplicity of message and the letter will be unnecessarily lengthy.
- Unnecessary words should not be used.
- Acronyms, abbreviations and jargon should also be avoided.
- Long paragraphs in a technical document are never preferred. In a long paragraph, a reader is forced to wade through many words and digest large amounts of information. Short paragraphs are easily managed. They invite reading and also help the reader to understand the content.
- One more thing which the writer should keep in mind is accuracy. It refers to a mistake free proof-reading.

Q.15. Comment on the vocabulary in technical writing.

Ans : A technical document differs from a general writing in terms of its vocabulary. As soon as a student opens a science book, he is able to understand clearly the basic difference between technical and general writing. A technical writing – say a book of science – uses many special words instead of the words that are used in day-to-day language. Thus, a book of science may use the words such as 'citric acid' for 'vinegar', the word 'apex' for 'top' and 'lateral' for 'slanting'. In the books of Botany, we find botanical names for common plants. Similarly, in general writing abstract words or phrases should not be used. Usually the familiar words should be used.

Q.16. Name the three levels of communication.

(B.Tech., 2008-2009)

Ans : Three levels of communication are :

Interpersonal, Organizational and Mass Communication.

Interpersonal Level of Communication : It involves two parties-a sender and a receiver who use common language to transit message either through oral communication or written communication.

Organizational Level of Communication : In a team-based business organization, communication becomes its lifeblood where people communicate with one another. The flow of communication inside an organization may filter in up, down and horizontal directions. Besides internal communication, companies depend on external communication also. Companies exchange messages with people outside the organization through external communication.

Mass Communication : Newspapers, magazines and periodicals, the means of mass communication, are frequently used for oral or written communication. Besides, technologies such as the internet, e-mail, voicemail, faxes, audiotape, teleconferencing, videoconferencing and closed circuit televisions have increased options for internal and external communication. These fast means help people from all parts of the world to work together.

Q.17. In what direction can information travel within an organization's formal hierarchy ?

Ans : In today's fast paced e-commerce environment, the command and control style of traditional management is no more effective. To function, in a team-based organization, it's necessary that information must travel down, up and across an organization's formal hierarchy i.e. if the managers or top officials convey decisions down to the co-workers, then the co-workers may also convey some worthwhile information up to their top officials. It's because a manager cannot be everywhere at once. Likewise communication in an organization must flow from one department to another horizontally. This attitude helps the employees to share information and coordinate task. Moreover, this horizontal or lateral communication may be helpful in some crises. Thus there can be three directions in either of which the information may flow :

Downward Flow : When information flows from top to bottom.

Upward Flow : When the message flows from bottom to top i.e. from the low-level workers to high officials.

Horizontal or Lateral Flow : When the message is conveyed from one department to another, among the people of the same rank.

Q.18. What is the grapevine ? Why should managers know how it works ?

Ans : There is an informal communication network in every organization. People talk to their friends unofficially in an office. Most of the time, their talks deal with personal matters but very often they casually discuss the official matters. This sort of conversation in an office is known as grapevine which may help the managers to spread and receive informal message, pertaining to business.

Q.19. What are the main barriers to communication ?

Ans : Communication can be successful only when the receiver understands the meaning of the message, sent by the sender. There are some communication barriers which may block the communication process and prevent the sender from transmitting his ideas meaningfully. A few communication barriers are recognized by the authors like Bovee et al :

- (i) **Perceptual and Language Difference :** Whenever two parties- the sender and the receiver-have different mental image or perception of an object, it leads to misunderstanding. Difference of language also plays a negative role and leads to confusion.
- (ii) **Restrictive Environment :** Whenever the people at the top prevent the lower-level employees from communicating their thoughts, such control and command attitude proves an imminent barrier to communication.
- (iii) **Distractions :** Some physical as well as emotional distractions may distort the message. Physical distractions like illegible⁵⁶ copy of the message, lack of proper lighting, health problems, lack of physical comforts may hinder communication. Likewise emotional distractions like the emotions of hate, anger, jealousy, fear etc. may also be a barrier in sending as well as in receiving a message.
- (iv) **Deceptive Tactics :** Use of deceptive tactics like exaggeration of the profits or concealment of the facts by the manipulative communicator may distort the message and lead to misunderstanding.
- (v) **Communication Network :** The information overloading also disrupts⁵⁷ the process of communication.

Q.20. Suggest the ways to overcome the communication barriers.

Ans : The main communication barriers can be overcome if the following guidelines are followed by the companies :

- (i) Adopt an audience based approach.
- (ii) Foster an open communication climate.
- (iii) Commit to ethical communication.
- (iv) Create lean and efficient messages.

Q.21. What is noise ?

Ans : Whenever there is any interference or break in the communication process, it makes difficult for the sender to convey his message and for the receiver to receive it. The message, thus, is partially conveyed or partially received, so will be the response. This is known as 'noise'.

Q.22. Discuss Language as a medium / tool of communication.

Ans : In order to express his ideas emphatically, man needs some common medium of expression. Physical gestures or certain noises like crying in anger, weeping or laughing may help a person to communicate some of his feelings, just as a little baby weeps when he is hungry or he slaps his mother when he is angry. In both the above examples, communication takes place without words. But man needs some means of linguistic⁵⁸ communication in order to express his fine ideas and emotions.

Man invented language which helps him to communicate and converse⁵⁹ with different sorts of people. Today it has become an important tool of socialization as well as of trade and commerce.

Q.23. Mention the three basic barriers to communication.

Ans : Three basic barriers to communication are:

- (i) **Intrapersonal**
 - (ii) **Interpersonal**
 - (iii) **Organizational**
- (i) **Intrapersonal Barriers :** This includes an individual's inbuilt barriers which may block the communication process. Some such causes are listed below :
- Different background -education, culture
 - Different perceptions and Language differences
 - Attitudes
 - Wrong assumptions
- (ii) **Interpersonal Barriers :** The barriers between person to person (i.e., the sender and the receiver) are termed as interpersonal barriers. These barriers also hinder the communication process. Some causes leading to interpersonal barriers are listed below :
- Emotional outbursts
 - Poor listening skill
 - Noises in the channel
 - Vocabulary
 - Deceptive tactics, adopted by the manipulative communicator

- (iii) **Organizational Barriers** : The communication barriers may exist in an entire organization which often disrupt the smooth flow of communication. Some causes of the organizational barriers are
- Restrictive environments where the style of management is very authoritarian.
 - Distractions
 - Choice of inappropriate media
 - Message overloading
 - Sense of insecurity : Fear of transfer

Q.24. What are the distinguishing features of technical communication ? Discuss organizational barriers to technical communication in a professional organization. (B. Tech., 2008-2009)

Ans : The term technical communication means the transmission of facts, figures, ideas and all sorts of scientific and technical information from one individual to another. Sharing information is actually the glue that binds the organization not only with the outside world but also within the organization. Technical communication which can be internal as well as external differs from general communication. Some salient features of technical communication are listed below. Technical communication is

- (i) specific and technical
- (ii) for specific audience
- (iii) formal in nature
- (iv) always objective
- (v) based on the principle of brevity and clarity

Sometimes some organizational barriers hinder the process of communication. Such organizational barriers are :

- (i) **Restrictive Environment** : "The communication climate suffers when information is distorted, fragmented or blocked by an authoritarian style of management", comment Bovee et al. Sometime the status relationships i.e., relationship between superior and subordinate creates hurdle in the effectiveness of communication.
- (ii) **Sense of Insecurity** : Lack of cooperation among the staff hinders the communication process. It becomes injurious when the fear of transfer or demotion prevents the subordinate staff to give suggestions or raise fingers to some malpractice in an organization.
- (iii) **Inadequate facilities** : Lack of adequate facilities of communications such as phone, e-mail, fax, no seminars and meetings also works as barriers to effective communication. In such a state, people choose inappropriate media that also hinder the communication process.
- (iv) **Rigid rules and regulations** : No doubt rules and regulations help in the smooth working of an organization. But sometime too rigid rules work as barriers to communication. For example, the complex hierarchical structure obstructs the free flow of communication because in such a system a junior staff member has no direct access to the top officials. Consequently when the message reaches the top official after passing through different other officials it either loses its basic form or is diluted.
- (v) **Distractions** : The distractions which intercept the message are generally physical or emotional. But sometimes these distractions arise due to the climate and culture of the organization where people find it difficult to adapt themselves to the need of the organizational set-up. Such distractions are then termed as organizational barriers.

Q.25. What are characteristic features of technical communication ? Why is simplicity more essential than sonorous and figurative style in technical writing ? Elucidate.

[B.Tech. SEM. II, 2010-11, M.T.U., Noida]

Ans : For answer, see answers to Q. numbers 12 & 13.

Q.26. What are intrapersonal barriers to communication ? Describe briefly.

[B.Tech. SEM. II 2010-11, B.Tech. SEM. I 2011-12, M.T.U., Noida]

Ans : An individual's inbuilt barriers which may block the communication process are called intrapersonal barriers to communication. Some such barriers which are listed below hinder the communication process and message does not reach the receiver due to such barriers taking place at his own level:

- Different education and culture background
- Different perceptions and language differences
- Attitudes
- Poor listening skills
- Wrong assumptions
- Emotional state
- Prejudgement

Q.27. Define communication.

[B.Tech. SEM. I 2011-12, M.T.U., Noida]

Ans : For answer, see answer to Q. number 1.

Q.28. Enumerate the merits and demerits of upward communication.

[B.Tech. SEM. I 2011-12, M.T.U., Noida]

Ans : Upward communication means communication from subordinates to superiors. Some merits of upward communication are :

- (i) It enables the top-officials to know what is going on in the organization.
- (ii) The subordinates begin to have faith in their superiors.
- (iii) The subordinates provide a feedback.
- (iv) Sharing information of an organization helps to achieve organizational goals.

There are some demerits also of this flow of communication.

- (i) The subordinates highlight only the positive points of a survey and neglect the negative points which may lead to some problems in the long way.
- (ii) The reluctance of the subordinates in communication turns into ineffectiveness of the upward flow of communication.
- (iii) Fear of the superiors and fear of demotion often prevents the subordinates from frank and free communication.

Q.29. Compare and contrast the downward flow of communication from the upward flow of communication.

[B.Tech. SEM. II 2012-13, G.B.T.U., Lucknow]

Ans : To function in a team based organization information flows from upward, downward and horizontal. Main decisions in an organization are usually taken by the top officials and then they flow down to the subordinates who will carry out them. In upward flow of communication, the co-workers submit their reports, proposals, requests, suggestion and some feed back to their superiors which helps the superiors to know about different problems, plans, complaints of the lower-level employees and work accordingly.

Q.30. What is technical communication ? Describe the features of brevity and utility precisely.

[B.Tech. SEM. II, 2012-13, G.B.T.U., Lucknow]

Ans : For answer, see answers to Q. numbers 6 & 12.

Q.31. What are the causes of interpersonal barriers ? How do poor listening skills affect interpersonal communication ?

[B.Tech. SEM. II, 2012-13, B.Tech. SEM. I 2012-13 G.B.T.U. Lucknow]

Ans : The barriers between the sender and the receiver are termed as interpersonal barriers. Some causes leading to interpersonal barriers are :

- ▶ Emotional outbursts
- ▶ Poor listening skill
- ▶ Noises in the channel
- ▶ Language
- ▶ Deceptive tactics of the manipulative communicator

Poor listening skill often hinders interpersonal communication. For successful communication, careful and attentive listening is essential. Listening is a skill and for ideal communication, it is necessary for the receiver to learn it. Listening also depends on the use of words, style of the communicator.

Q.32. What are basic elements of the process of communication ? Enlist the most important levels of communication.

[B.Tech. SEM. II, 2011-12, G.B.T.U., Lucknow]

Ans : For answer, see answers to Q. numbers 3 & 16.

Q.33. Discuss the origin of technical writing. How could technical writing be distinguished from general writing ?

[B.Tech. SEM. II, 2011-12, G.B.T.U., Lucknow]

Ans : The efforts of Royal society in 1660 promoted the scientific researches. Later the out-break of the industrial revolution in 18th century brought revolution in the field of technical communication. Today, technologies like electronic mail, video conferencing, instant messaging and chatting, internet telephony file transfers etc. are possible due to technological advancement. All these are known as parts of technical writing. (For rest of the answer see answer to Q. number 5).

Objective Type Questions

Directions: Choose correct answer/most appropriate answer for each one:

1. What does technical communication stand for ?

- (a) Communication for technicians
- (b) Communication for laymen
- (c) General communication
- (d) Communication for professionals

(B.Tech., 2008-09)

2. What does general communication mean ?

- (a) General science communication
- (b) Communication for military generals
- (c) Personal and mass communication
- (d) None of the above

(B.Tech., 2008-09)

3. What is downward communication ?

- (a) Communication with downtrodden
- (b) Fallen communication
- (c) From junior staff to senior staff
- (d) From top management to subordinate staff.

(B.Tech., 2008-09)

4. What are the purposes of technical communication ?

- (a) Exchange of information
- (b) Persuasion
- (c) Provide instructions
- (d) All the above

5. Communication carried with a large audience is known as

- (a) Mass communication
- (b) Verbal communication
- (c) Non-verbal communication
- (d) Interpersonal communication

6. Horizontal or lateral communication takes place:

- (a) From top to bottom
- (b) From bottom to top
- (c) Among the peer group.
- (d) None of the above

7. The reciprocal process of communication is known as

- (a) One way communication
- (b) Two way communication
- (c) Mass communication
- (d) Symbolic communication.

8. Before the sender sends a message, he
 - (a) Encodes the message
 - (b) Decodes the message
 - (c) Transmits the message
 - (d) Sends the feedback.
9. Main levels of communication are
 - (a) Interpersonal
 - (b) Organizational
 - (c) Mass Communication
 - (d) All the above
10. What is Intra-personal barrier to communication ? (2008-2009)
 - (a) Barrier between person to person
 - (b) Individuals inbuilt barriers
 - (c) External barriers
 - (d) Emotional barriers

(B.Tech., 2008-09)
11. What is inter-personal barrier to communication ?
 - (a) Enmity between persons
 - (b) Barrier of selfishness
 - (c) Barrier of mutual differences
 - (d) Barrier caused by inappropriate transaction of words between two or more

(M.C.A., 2008-09)
12. Main features of general communication are
 - (a) To amuse the audience/readers.
 - (b) Cast in decorative style.
 - (c) Objective in style
 - (d) Accurate and precise.
13. Main gateways to communication are
 - (a) Audience-centred approach
 - (b) Open communication climate
 - (c) Ethical communication
 - (d) All the above
14. What is noise?
 - (a) Interference in the communication process
 - (b) Mixture of loud, unpleasant sounds
 - (c) Use of indecent language
 - (d) Use of unintelligible sounds
15. Technical communication stands for which of the following ?
 - (a) It is communication of technicians
 - (b) It is communication of mechanics
 - (c) It is communication of professionals on technical usage.

[B.Tech. SEM. II 2010-11 M.T.U., Noida]

16. Interpersonal communication involves :
 - (a) Self-communication
 - (b) Communication with another person
 - (c) Communication of the individual with many persons.

[B.Tech. SEM. II, 2010-11, M.T.U., Noida]
17. Downward Communication is meant for which of the following ?
 - (a) With downtrodden people
 - (b) Flows from head of the organization down to the lowest in the official command
 - (c) Low-level people/employees.

[B.Tech. SEM. II, 2010-11, M.T.U., Noida]
18. Intrapersonal barriers to communication are caused by which of the following ?
 - (a) Personal inbuilt traits/situations of the individual
 - (b) Deficiencies/shortcomings of other persons
 - (c) Deficiency/shortcoming of one person with other person.

[B.Tech. SEM. II, 2010-11, M.T.U., Noida]
19. Limited vocabulary affects which type of communication ?
 - (a) Interpersonal communication
 - (b) Intrapersonal communication
 - (c) Official communication

[B.Tech. SEM. II, 2010-11, M.T.U., Noida]
20. Fear of superiors is related to which of the following ?
 - (a) Intrapersonal barriers to communication
 - (b) Organizational barriers to communication
 - (c) Interpersonal barriers to communication

[B.Tech. SEM. II, 2010-11, M.T.U., Noida]

Directions: Match the parts of List A with that of List B :

(a) Communication is a	spontaneous expression of responses/ideas
(b) Communication between higher to lower level	an event carried back to some one
(c) Informal communication involves	are called imperative sentences
(d) Sentences which command or request actions	is downward communication
(e) A report is a description of	circular process

[B.Tech. SEM. I 2009-10, G.B.T.U., Lucknow]

Directions: Fill in the Blanks in the following questions :

1. The directions of communication can be downward, and horizontal.
2. Communication can be formal and
3. The informal channel of communication is often called the
4. of messages often hinder the communication process.
5. Break in the communication process means
6. An individual's inbuilt barriers are known as
7. Under upward communication the flow of information is from level.

Directions: Tell whether the following statements are True or False :

1. There is no essential difference between technical communication and general communication.
2. Communication is defined as an activity or process of expressing ideas and feelings or of giving people information.
3. Technical communication is for specific audience whereas general communication is for all.
4. The receiver encodes the message.
5. External communication carries information into and out of the organization.
6. Open communication climate facilitates the communication process in an organization.

Multiple Choice Questions

1. (d)	2. (c)	3. (d)	4. (d)	5. (a)	6. (c)	7. (b)
8. (a)	9. (d)	10. (b)	11. (c)	12. (a)	13. (d)	14. (a)
15. (c)	16. (b)	17. (b)	18. (a)	19. (c)	20. (b)	

Match the Following

- (a) circular process
- (b) is downward communication
- (c) spontaneous expression of responses/ideas
- (d) are called imperative sentences
- (e) an event carried back to some one

Fill in the Blanks

1. upward	2. informal	3. grapevine	4. over loading
5. noise	6. intrapersonal barriers	7. lower	

True or False

1. False	2. True	3. True
4. False	5. True	6. True

Things To Remember

What is Communication?

Communication is "the activity or process of expressing ideas and feelings or of giving people information".

Language is a medium of communication.

What is Technical Communication?

Technical communication means the transmission of facts, figures, ideas and all sorts of scientific and technical information from one individual to another.

All communication, general or technical, involves two persons—a speaker or a writer who encodes a concept which he wants to communicate and a listener or a reader who decodes the message and converts it into meaning.

Origin and Scope of Technical Communication

The efforts of Royal Society in 1660 promoted the scientific researches. Later the out-break of the industrial revolution in 18th century brought revolution in the field of technical communication. Today, technologies such as the internet, e-mail, voice mail and faxes allow people to work faster, better and also in cheaper way. It paves the path for global collaboration.

What is Technical Written Communication ?

"Technical writing is the practical writing that people do as part of their jobs", write Pauley and Riordan. Different kinds of documents such as reports, proposals, technical articles, instruction manuals, brochures etc. are the means of technical written communication.

Impact of Industrial Revolution on Technical Writing

Electronic mail, video conferencing, instant messaging and chat, internet telephony file transfers, locate information from external sources, all this global collaboration is possible due to these gifts of industrial revolution.

Significance/ Relevance of Technical Writing

Technical Writing is used either

- To inform
- To instruct
- To persuade

Features of Technical Writings

1. Use of familiar and easy words.
2. Use of clear sentences.
3. Use of sentences of moderate length.
4. Use of active voice.
5. Avoid redundant phrases.
6. Avoid inaccurate expressions.
7. Logical division of paragraph.

Features of General Writing

1. Content may be general in nature.
2. Purpose may be to amuse the readers.
3. General writings are often cast in personal or subjective style.
4. They are cast in decorative style, including use of bombastic words.
5. They are not meant for a specific reader.

Difference between Technical Writings and General Writings

Technical Writings	General Writings
1. Aim is to impart information, instructions and explanations	1. Aims at amusement
2. Objective Style	2. Subjective/ personal style
3. Accurate and precise	3. Decorative and bombastic
4. Simple but effective	4. Use of extravagant words
5. Related to specific profession	5. Related to life in general
6. Use of technical terms.	6. Use of common words

Style: Objective Style as Contrary to Literary Composition

Objective Style	Literary Composition
1. Precise and accurate	1. Use of superfluous words
2. Simple and clear	2. Use of figurative language
3. Content is impersonal	3. Content is personal
4. Use of technical terms	4. Use of common terms

Style in writing depends on

- content
- the audience and
- the purpose

Language as a Tool of Communication

- Helps in transmitting ideas and messages from one person to another.
- The sender and the receiver should have a common language

Levels of Communication

Interpersonal : Communication between a sender and a receiver.

Organizational : Communication in and out of organization and also among employees of an organization serving on different posts.

Mass communication : Fast communication through newspapers, magazines etc. depending on modern technological equipments.

The Flow of Communication

Downward : From the top to bottom.

Upward : From the bottom to top.

Horizontal or lateral : Among the peer group.

Barriers to Communication

- Perceptual and Language Differences.
- Restrictive Environment.
- Distractions : physical as well as emotional.
- Deceptive Tactics.
- Communication Network : Information overloading.

Another way of Categorizing Communication Barriers

- Physical Barriers
 - ▶ Noises
 - ▶ Environment
 - ▶ Defects of Channel, Medium and Media
 - ▶ Inadequate Facilities

- **Semantic Barriers**
- **Organizational Barriers**
 - ▶ Restrictive Environments
 - ▶ Deceptive Tactics
 - ▶ Sense of Insecurity
 - ▶ Rigid Rules and Regulations
 - ▶ Closed Communication Climate
- **Socio-Psychological Barriers**
 - ▶ Attitudes
 - ▶ Emotions/Moods
 - ▶ Resistance to Change

Ways to Overcome the Communication Barriers

- Adopt an audience centred approach.
- Foster an open communication climate.
- Commit to Ethical communication.
- Create lean and efficient messages.